

Kenneth Chen

PhD Student

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Research Interests

I am broadly interested in computer graphics & displays, VR/AR, psychophysics, high dynamic range, color, image quality, applications of AI, and more. My thesis work focuses on developing algorithms for optimizing immersive displays, especially by leveraging data from studies of human perception and visual psychophysics.

Education

New York University

2023 – 2027

PhD in Computer Science

2021 – 2023

MS in Computer Science

Advisor: Qi Sun

Committee: Daniele Panozzo, Claudio Silva, Rafal K. Mantiuk, Aaron Hertzmann

University of California, Berkeley

2017 – 2021

BA in Computer Science

Advised by Prof. Carlo H. Séquin

Swim Club @ Berkeley, Innovative Design @ Cal

Publications

2026

Adapting Quality Metrics to Tone Mapping

K. Chen, D. Kim, Y. Asano, A. Chapiro, Q. Sun, R. Mantiuk

ACM SIGGRAPH

Dichoptic Foveation

H. Kam, C. Groth, J. Kang, P. Saraf, Q. Sun, **K. Chen**

ACM SIGGRAPH

Dynamic Visual Dominance in Stereoscopic Foveation

D. Jiménez Navarro, C. Groth, **K. Chen**, Q. Sun, K. Myszkowski, H. Seidel, A. Serrano

ACM SIGGRAPH

Cost-Aware Routing for Efficient Text-To-Image Generation

Q. Li, **K. Chen**, C. Su, W. Jitkrittum, , Q. Sun, P. Sangkloy

Transactions on Machine Learning Research (TMLR)

ML-PEA: Machine Learning-Based Perceptual Algorithms for Display Power Optimization

K. Chen, N. Matsuda, T. Wan, A. Ninan, A. Chapiro, Q. Sun

Eurographics (Proc. Computer Graphics Forum)

Perceptual Impact of Peak Luminance and Contrast in Direct View HDR Display

K. Chen^{*}, Y. Zhang^{*}, Q. Sun, A. Chapiro

IS&T Electronic Imaging (EI)

Perceptually Guided 3DGS Streaming and Rendering for Virtual Reality

Y. Zhang, S. Mupparaju, K. Chen, J. Kang, X. Zhang, M. Omori, K. Arimatsu, Q. Sun

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

GeneVA: A Dataset of Human Annotations for Generative Text to Video Artifacts

J. Kang, P. Sangkloy, M. Silva, K. Chen, N. Williams, Q. Sun

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)

2025

What is HDR? Perceptual Impact of Luminance and Contrast in Immersive Displays

K. Chen, N. Matsuda, J. McElvain, Y. Zhao, T. Wan, Q. Sun[†], A. Chapiro[†]

ACM SIGGRAPH

Image-GS: Content-Adaptive Image Representation via 2D Gaussians

Y. Zhang, B. Li, A. Kuznetsov, A. Jindal, S. Diolatzis, K. Chen, A. Sochenov, A. Kaplanyan, Q. Sun

ACM SIGGRAPH

Novel View Synthesis for 3D Computer-Generated Holograms Using Deep Neural Fields

K. Chen, A. Wen, Y. Zhang, P. Chakravarthula, Q. Sun

Optics Express

Process Only Where You Look: Hardware and Algorithm Co-optimization for Efficient Gaze-Tracked Image Rendering in Virtual Reality

H. Wang, W. Liu, K. Chen, Q. Sun, S. Q. Zhang

ACM/IEEE International Symposium on Computer Architecture (ISCA)

Perceptually-Guided Acoustic “Foveation”

X. Peng, K. Chen, I. Roman, J. P. Bello, P. Chakravarthula, Q. Sun

IEEE Conference on Virtual Reality and 3D User Interfaces (VR)

2024

PEA-PODs: Perceptual Evaluation of Algorithms for Power Optimization in XR Displays

K. Chen, T. Wan, N. Matsuda, A. Ninan, A. Chapiro[†], Q. Sun[†]

ACM SIGGRAPH (Proc. ACM ToG) | Best Paper Award, Honorable Mention

Exploiting Human Color Discrimination for Memory- and Energy-Efficient Image Encoding in Virtual Reality

N. Ujjainkar, E. Shahan, K. Chen, B. Duinkharjav, Q. Sun, Y. Zhu

ACM Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)

2023

Toward Optimized VR/AR Ergonomics: Modeling and Predicting User Neck Muscle Contraction

Y. Zhang, K. Chen, Q. Sun

ACM SIGGRAPH

2022 **Color-Perception-Guided Display Power Reduction for Virtual Reality**

B. Duinkharjav*, **K. Chen** *, A. Tyagi, J. He, Y. Zhu, Q. Sun

ACM SIGGRAPH Asia (Proc. ACM ToG)

Other Publications (Demos, Posters, etc.)

2026 **Limits of Visual Saliency Models for AI-Generated Videos**

J. Kang, N. Williams, M. Silva, **K. Chen** , P. Sangkloy, Q. Sun

Vision Sciences Society (Applied Vision Science meeting) | Poster

2024 **BlendFusion: Procedural 3D Texturing Assistant with View-Consistent Generative Models**

Q. Li, F. Torrens, **K. Chen** , Q. Sun

ACM Symposium on Interactive 3D Graphics and Games (I3D) | Poster | **Best Poster Award**

2023 **Imperceptible Color Modulation for Power Saving in VR/AR**

K. Chen , B. Duinkharjav, N. Ujjainkar, E. Shahan, A. Tyagi, J. He, Y. Zhu, Q. Sun

ACM SIGGRAPH | Emerging Technologies

Towards Learning and Generating Audience Motion from Video

K. Chen , N. I. Badler

ACM Symposium on Computer Animation (SCA) | Poster

Honors and Awards

2025 **Top 10 Fast Forward, Audience Choice, ACM SIGGRAPH 2025**

For “What is HDR? Perceptual Impact of Luminance and Contrast in Immersive Displays”.

2025 **Best Poster Award, ACM I3D 2025**

For “BlendFusion: Procedural 3D Texturing Assistant with View-Consistent Generative Models”.

2025 **Deborah Rosenthal, MD Award, NYU**

“Awarded to a graduate student in CS for outstanding performance on the PhD qualifying examination.”

2024 – 2025 **U.S. DoE Graduate Assistance in Areas of National Need Fellowship (GAANN), NYU**

“This program provides fellowships, through academic departments and programs of IHEs, to assist graduate students with excellent records who demonstrate financial need and plan to pursue the highest degree available in their course study at the institution in a field designated as an area of national need.”

2024 **Best Paper Award (Honorable Mention), ACM SIGGRAPH 2024**

For the paper “PEA-PODs: Perceptual Evaluation of Algorithms for Power Optimization in XR Displays”.

2023 **Tandon School of Engineering PhD Fellowship, NYU**

2021 **Graduate Scholarship, NYU**

2020 **Computer Graphics (CS 184) Final Project Showcase Winner, UC Berkeley**

2019 **Probability and Random Processes (EE 126) Catan AI Competition 1st Place, UC Berkeley**

Patents

- 2025 **Perceptual Algorithms and Design Interface to Save Display Power**
A. Chapiro, **K. Chen**, A. Ninan, R. Bachy, J. Hillis, C. C. Wan, N. Matsuda
US Patent 2025 0199750 A1
- 2025 **Color-Perception-Guided Display Power Reduction for Extended Reality**
Q. Sun, Y. Zhu, B. Duinkharjav, **K. Chen**, A. Tyagi, J. He
US Patent 2025 0095532 A1
- 2025 **Human Color Discrimination for Memory-Efficient Encoding in Extended Reality**
Y. Zhu, Q. Sun, N. Ujjainkar, E. Shahan, **K. Chen**, B. Duinkharjav
US Patent 2025 0095232 A1

Research Experience

- 6/'26 - 8/'26 **Adobe Research** | Research Scientist Intern
- 6/'25 - 8/'25 **University of Cambridge** | Visiting Student
w/ Prof. Rafal Mantiuk
Worked on image/video quality metrics for tone-mapped content.
Resulted in a SIGGRAPH 2026 publication.
- 6/'24 - 1/'25 **Meta Reality Labs, Applied Perception Science** | Research Scientist Intern
w/ Dr. Alexandre Chapiro
Studied/modelled contrast and luminance preferences in HDR displays.
Resulted in a SIGGRAPH 2025 publication.
- 6/'23 - 1/'24 **Meta Reality Labs, Applied Perception Science** | Research Scientist Intern
w/ Dr. Alexandre Chapiro
Studied the visibility of different display power optimization algorithms in a VR setting.
Measured and modeled power consumption of the Quest Pro display.
Resulted in a SIGGRAPH 2024 paper which won a best paper award, honorable mention.
- 6/'22 - 8/'22 **Cesium GS** | Research Scientist Intern
w/ Prof. Norman I. Badler
Resulted in a poster presentation at SCA 2023.
- 9/'18 - 5/'19 **UC Berkeley** | Undergraduate Research Assistant
w/ Prof. Carlo H. Séquin

Teaching

- SP 2025 **Data Visualization (CS-GY 6313)**
New York University, Course Assistant
- FA 2024 **Virtual and Augmented Reality (CS-GY 9223)**
New York University, Course Assistant
- SP 2023 **Data Visualization (CS-GY 6313)**
New York University, Course Assistant
- FA 2021 **Interactive Computer Graphics (CS-GY 6533)**

New York University, Course Assistant

SP 2021 **Computer Graphics and Imaging (CS 184)**
UC Berkeley, Undergraduate Student Instructor

FA 2020 **Introduction to Computer Vision and Computational Photography (CS 194-26)**
UC Berkeley, Group Tutor & Reader

Talks

05/2026 **ML-PEA: Machine Learning-Based Perceptual Algorithms for Display Power Optimization**
Eurographics | Aachen, Germany

Power-Efficient Immersive Perceptual Graphics & Displays

05/2026 Eurographics Doctoral Consortium | Aachen, Germany

03/2026 Physical AI Journal Club | Stanford University
Hosted by Prof. Gordon Wetzstein

03/2026 **Perceptual Impact of Peak Luminance and Contrast in Direct View HDR Display**
IS&T HVEI 2026 | San Francisco, California

08/2025 **What is HDR? Perceptual Impact of Luminance and Contrast in Immersive Displays**
ACM SIGGRAPH 2025 | Vancouver, Canada

Perceptually-Inspired Algorithms for Power Optimization in XR Displays

07/2025 Rainbow Seminar | University of Cambridge
Invited by Prof. Rafal Mantiuk

11/2024 High-Beams Seminar | Virtual
Invited by Prof. Kaan Aksit

08/2024 **PEAPODs: Perceptual Evaluation of Algorithms for Power Optimization in XR Displays**
ACM SIGGRAPH 2024 | Denver, Colorado

12/2022 **Color-Perception-Guided Display Power Reduction for Virtual Reality**
ACM SIGGRAPH Asia 2022 | Daegu, South Korea

Organizing

07/2026 [From Human Vision to Machine Vision Workshop](#)
K. Chen, N. L. Williams, C. Groth, J. Kang, A. Chapiro, Q. Sun
ACM SIGGRAPH | Technical Workshop

03/2026 [Asian Americans & STEM Conference](#)

03/2026 [Workshop on Perception-driven Graphics and Displays for VR and AR \(PerGraVAR\)](#)
M. Weier, O. Staadt, C. Groth, B. Mohanto, A. Marquardt, C. Salaün, P. Chakravarthula, D. Martin, **K. Chen**
IEEE Conference on Virtual Reality and 3D User Interfaces (VR) | Workshop

Academic Service

Program Committee:

ACM IUI | 2026, 2025

Peer Review:

IEEE/CVF CVPR | 2026

IEEE/CVF WACV | 2026

Eurographics | 2025

ACM IUI | 2026, 2025

ACM CHI | 2026

IEEE VR | 2026, 2025

ACM ToG | 2025

EuroXR | 2025

ACM SIGGRAPH | 2026, 2025, 2024

ACM SIGGRAPH Asia | 2025, 2024

Computers and Graphics | 2026

IEEE TVCG | 2025, 2024

Mentorship

2023 – 2026	Henry Kam, <i>BS CS at NYU</i> <i>Resulted in a SIGGRAPH 2026 paper.</i>
2025	Maria Beatriz Silva, <i>BS CS at NYU</i> <i>Resulted in a WACV 2026 paper.</i>
2025	Harsha Mupparaju, <i>MS CS at NYU</i> <i>Resulted in a WACV 2026 paper.</i>
2024 – 2025	Qinchao Li, <i>MS CS at NYU (Now PhD Student @ SFU)</i> <i>Resulted in an I3D 2025 poster which won a best poster award.</i>
2024 – 2025	Changyue (Tina) Su, <i>MS CS at NYU (Now SDE @ Meta)</i> <i>Resulted in a TMLR 2026 paper.</i>
2023	Luigia Than, <i>MS CE at NYU</i>

Outreach & Volunteering

2026	ACM SIGGRAPH History and Archive Committee <i>Volunteer</i>
2020 – 2021	Innovative Design at Cal <i>Vice President of Technology</i> Led a web development team, provided pro-bono technical services to on-campus orgs.
2018	Special Needs Aquatics Program (SNAP kids) <i>Volunteer Instructor</i> Swim lessons and exercises with special needs/limited mobility children.

Media Coverage

2025	Two Minute Papers Intel Just Changed Computer Graphics Forever!
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- 2025 **Intel Blogs** | [Visual Efficiency for Intel's GPUs](#)
- 2024 **NYU Engineering News** | [Immersive Computing Lab teams up with Meta to uncover how energy-saving tactics affect perceived quality of XR experiences.](#)

Other Employment

- FA 2021 **NVIDIA** | Software Engineer Intern
Developed C++ Omniverse plugin for NVIDIA DRIVE Sim.
- SU 2021 **Analog Devices** | Software Engineer Intern
Worked on point cloud rendering.
- SU 2020 **Confluent** | Software Engineer Intern
Fronted engineering for data governance.
- SU 2019 **Intuit** | Software Engineer Intern
Developed frontend tax widget for QuickBooks Payroll.
- SU 2016 **AMS Pools** | Head Lifeguard
Managed a neighborhood swimming pool; cleaning, checking/balancing chemicals.